

Topic: Evolution of Computers -- Self-Learning

Q. Trace the evolution of computers through their five generations. Highlight the key technological advancements and characteristics of each generation.

[DEF] Definition to Remember

The Evolution of Computers is divided into five "Generations", each marked by a major technological breakthrough that made computers smaller, cheaper, faster, and more reliable. The progression moved from Vacuum Tubes -> Transistors -> Integrated Circuits -> Microprocessors -> Artificial Intelligence.

Evolution of Computers (Self-Learning)

Generation	Core Technology	Key Characteristics & Examples
First (1940s - 1950s)	Vacuum Tubes (circuitry) & Magneti	Massive (room-sized), high heat, high
Second (1950s - 1960s)	Transistors	Smaller, faster, more energy-efficient
Third (1960s - 1970s)	Integrated Circuits (ICs)	Miniaturized transistors placed on sil
Fourth (1970s - Present)	Microprocessors (VLSI)	Thousands of ICs on a single silicon c
Fifth (Present & Beyond)	Artificial Intelligence (AI)	Based on parallel processing and Machi

[MUST-WRITE] Must-Write Points (for 10 marks)

1. Computer evolution is categorized into Five Generations.
2. 1st Gen: Used Vacuum tubes; massive, generated high heat, used machine language (ENIAC).
3. 2nd Gen: Replaced tubes with Transistors; smaller/faster, used assembly and high-level languages (COBOL).
4. 3rd Gen: Used Integrated Circuits (ICs) on silicon chips; introduced OS, keyboards, monitors.
5. 4th Gen: Used Microprocessors (VLSI); brought computers to desktops, introduced networking/Internet.
6. 5th Gen: Focused on Artificial Intelligence (AI) and parallel processing; natural language understanding.
7. The consistent trend across generations: Smaller size, lower cost, higher speed, and greater reliability.

[RECALL] Quick Recall

Generations: 1st (Vacuum Tubes, Room size) -> 2nd (Transistors, Assembly Lang) -> 3rd (Integrated Circuits, OS) -> 4th (Microprocessors/VLSI, PCs/Internet) -> 5th (AI, Machine Learning)